Jacek Karwowski

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EDUCATION

Ph.D. in Computer Science *2022 - now*
University of Oxford
Specialization: Probability theory and category theory

M.Sc. in Mathematics *2018 - 2021*
University of Warsaw
Thesis: Formal semantics of a reversible language

B.Sc. in Computer Science *2018 - 2021*
University of Warsaw
Thesis: Low-latency Apache Parquet library (blogpost)

B.Sc. in Mathematics *2015 - 2018*
University of Warsaw
Thesis: Products in positive opetopic sets

PUBLICATIONS

V. Choudhury, J. Karwowski, and A. Sabry. Symmetries in reversible programming: from symmetric rig groupoids to reversible programming languages. *Proc. ACM Program. Lang.*, 6(POPL), Jan. 2022

N. Ackerman, C. E. Freer, Y. Kaddar, J. Karwowski, S. Moss, D. Roy, S. Staton, and H. Yang. Probabilistic programming interfaces for random graphs: Markov categories, graphons, and nominal sets. *Proc. ACM Program. Lang.*, 8(POPL), Jan. 2024

R. Douglas, J. Karwowski, C. Bae, A. Draguns, and V. Krakovna. Limitations of agents simulated by predictive models. In *ICLR 2024 workshop on large language model (LLM) agents*, 2024

J. Karwowski, O. Hayman, X. Bai, K. Kiendlhofer, C. Griffin, and J. M. V. Skalse. Goodhart's law in reinforcement learning. In *The twelfth international conference on learning representations, ICLR*, 2024

J. Janiak, J. Karwowski, C. S. Mangat, G. Giglemiani, N. Petrova, and S. Heimersheim. Characterizing stable regions in the residual stream of LLMs. In *NeurIPS 2024 workshop on scientific methods for understanding deep learning*, 2024

TEACHING & ACADEMIC

Teaching

- **Teaching Assistant** - Probability Theory, University of Oxford 2024
- **Teaching Assistant** - Bayesian Statistical Probabilistic Programming, University of Oxford 2024
- **Teaching Assistant** - Lambda Calculus and Types, University of Oxford 2023
- **Teaching Assistant** - Introduction to AI Alignment, Oxford AI Safety 2023
- **Grader** - Linear Algebra, University of Warsaw 2021

Academic Service

- **Reviewer** - ICLR 2025, NeurIPS 2024 (top reviewer), CAV 2023
- **Program Officer** - PL in ML Conference 2017

Scholarships & Fellowships

- **Research Scholarship** - SERI ML Alignment & Theory Scholars 2023
- **PhD Scholarship** - Improving the Long-Term Future, Open Philanthropy 2022 - 2025
- **Fellowship** - Global Talent Attraction Program, Indiana University 2019

WORK EXPERIENCE

Researcher - National Institute of Informatics 2025

I worked on the information-geometric aspects of probabilistic programming languages theory at Sugiyama Lab, National Institute of Informatics in Tokyo.

Research lead - Supervised Program for Alignment Research 2025 - ongoing

I supervised a project *"Understanding and verifying the autoregressive conditioning hypothesis"* (see the description). The team consisted of six people with backgrounds in computer science and mathematics.

Researcher - Stanford Existential Risks Initiative 2023-2023

As part of SERI ML Alignment Theory Scholars program, I worked with Victoria Krakovna and others on understanding how can powerseeking arise in predictive models.

Researcher - Oxford AI Safety Labs 2022 - 2023

I worked with Joar Skalske at the Future of Humanity Institute and others on understanding Goodhart's law in reinforcement learning.

Quantitative Developer - Xantium 2021 - 2022

Xantium is the algorithmic trading division of Tudor Corporation. I worked on volatility trading, where I built models, implemented execution and managed datasets. Work details under NDA.

SWE Intern - G-Research 2020

G-Research is a quantitative research company, developing algorithms for predicting movements in financial markets. I worked on programmers tools for the simulations platform. Work details under NDA.

Research Intern - Indiana University 2019

As part of Global Talent Attraction Program, I conducted research on understanding the semantics of reversible programming languages, in the context of Homotopy Type Theory.

SWE Intern - Microsoft 2019

I worked on Office365 back-end, adding new features and optimizing the existing ones, and deploying code to production. Work details under NDA.

Research Intern - Golem Factory GmbH 2017 - 2018

Golem is a blockchain startup, creating a platform for sharing computational resources using Ethereum network. I developed a probabilistic verification algorithm for programs executed in an untrusted environment, and implemented it for the use-case of hyper-parameter optimization for neural networks (whitepaper, blogpost, another blogpost).

Data Scientist - Applica.ai 2016 - 2017

Applica is a startup working on ML tools for NLP. I was responsible for creating new models: from reading papers and designing experiments to prototyping models in Python and implementing them on the production cluster.

SWE Intern - University College London 2014 - 2015

Over the academic year, I have worked on developing a platform for testing high-frequency trading algorithms, and on an open-source medical data platform.